POWER BI

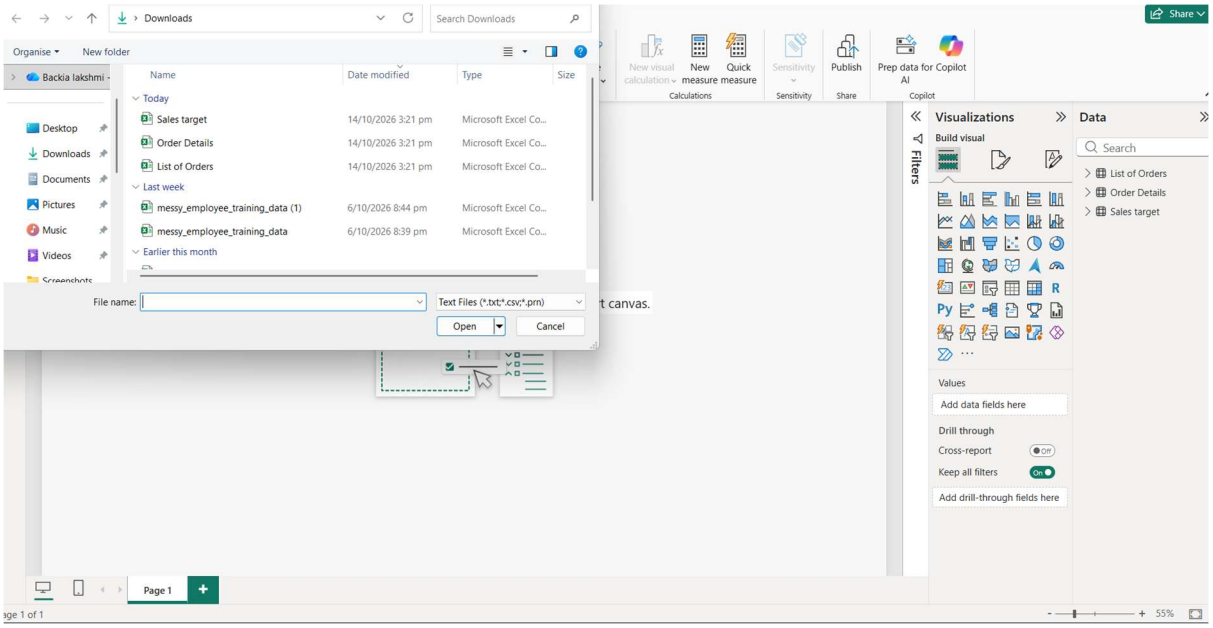
DATA TRANSFORMATION AND DATA MODELLING

DATASET DESCRIPTION :

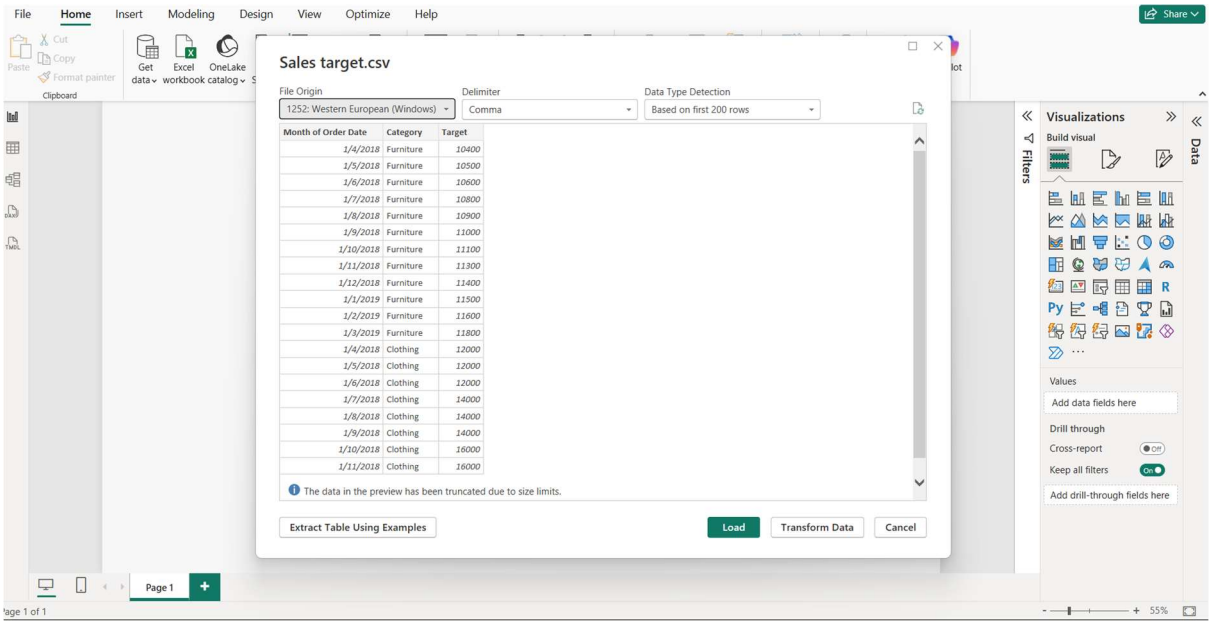
Order Detiles dataset contains order-level sales information, including order IDs, sales amounts, profit, quantities sold, product categories, and sub-categories. It helps analyze sales performance, profitability, product demand, and category-wise business results.Sales Target dataset contains monthly sales targets for different product categories, including the order month, category, and target value. It helps compare actual sales with target sales and identify whether each category meets its monthly goals.List of orders dataset for order id,date and customer name and also indicate the Location.

COLUMN	DESCRIPTION
Order ID	Unique identifier assigned to an order.
Amount	Monetary value of the sales transaction.
Profit	Profit or loss earned from the transaction.
Quantity	Number of units sold in the transaction.
Category	Main product category, such as Furniture, Clothing, or Electronics.
Sub-Category	More specific classification of a product within its main category.
Month of Order Date	Month and year for which the sales target is defined.
Category	Product category for which the target is set.
Target	Planned sales amount for the specified month and category.
Location	Cutomers easily identified.

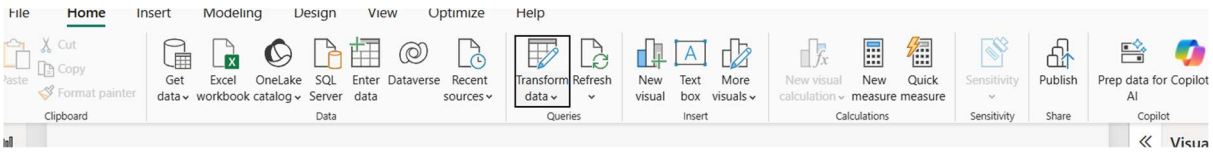
- Import List of Orders.csv into Power BI.

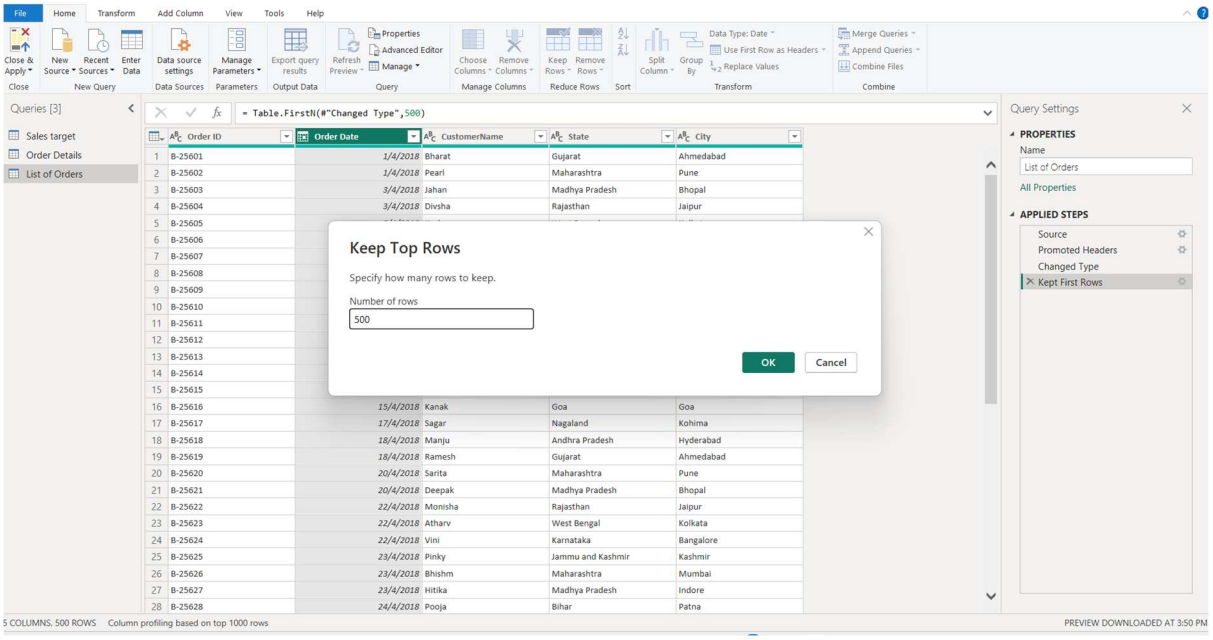


- Open the list of orders in the Power Query Editor using the Transform Data option.
- Import Order Details.csv and Sales Target.csv into Power Query Editor.

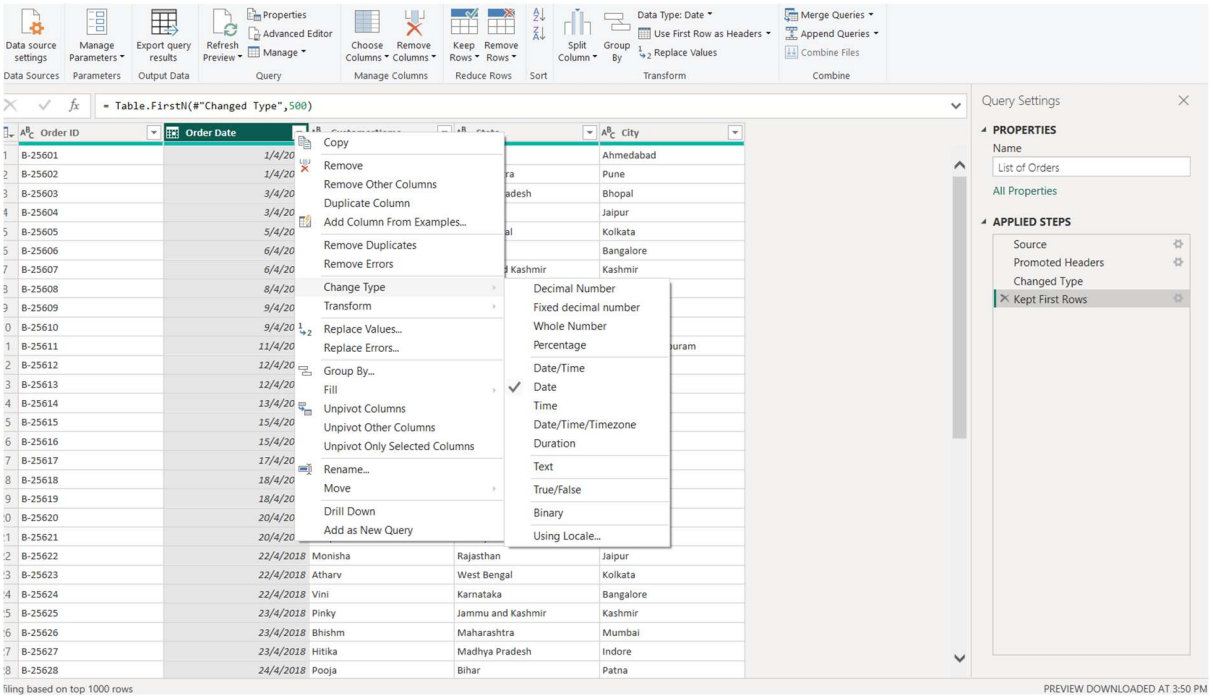


- Restrict the List of Orders table to the first 500 rows.

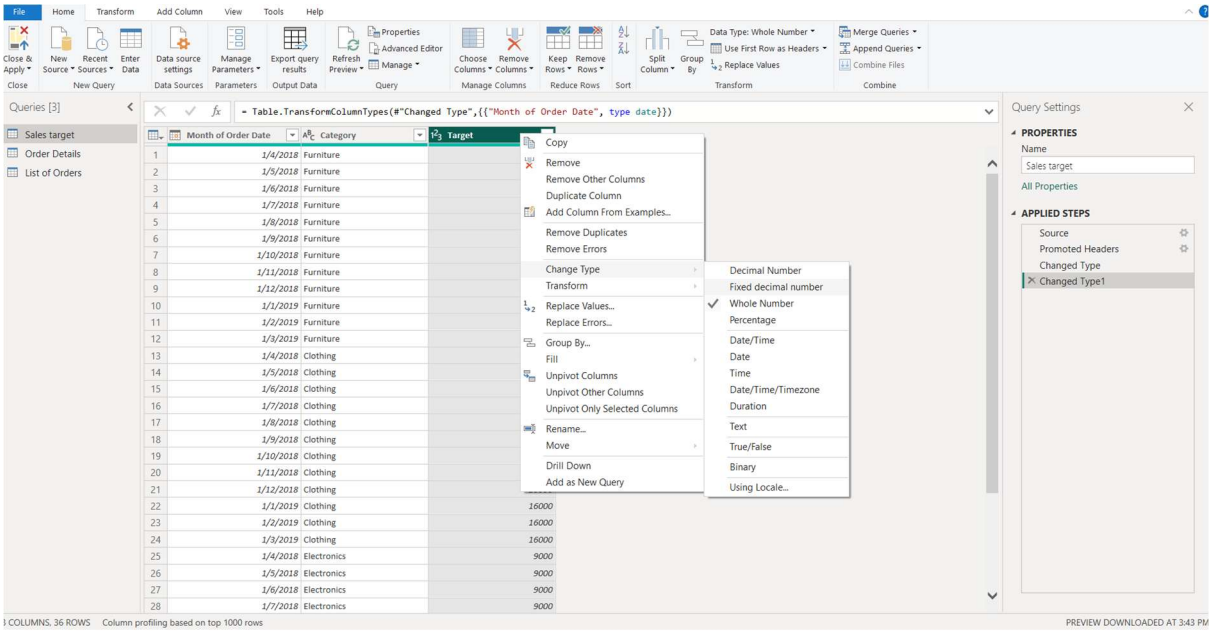


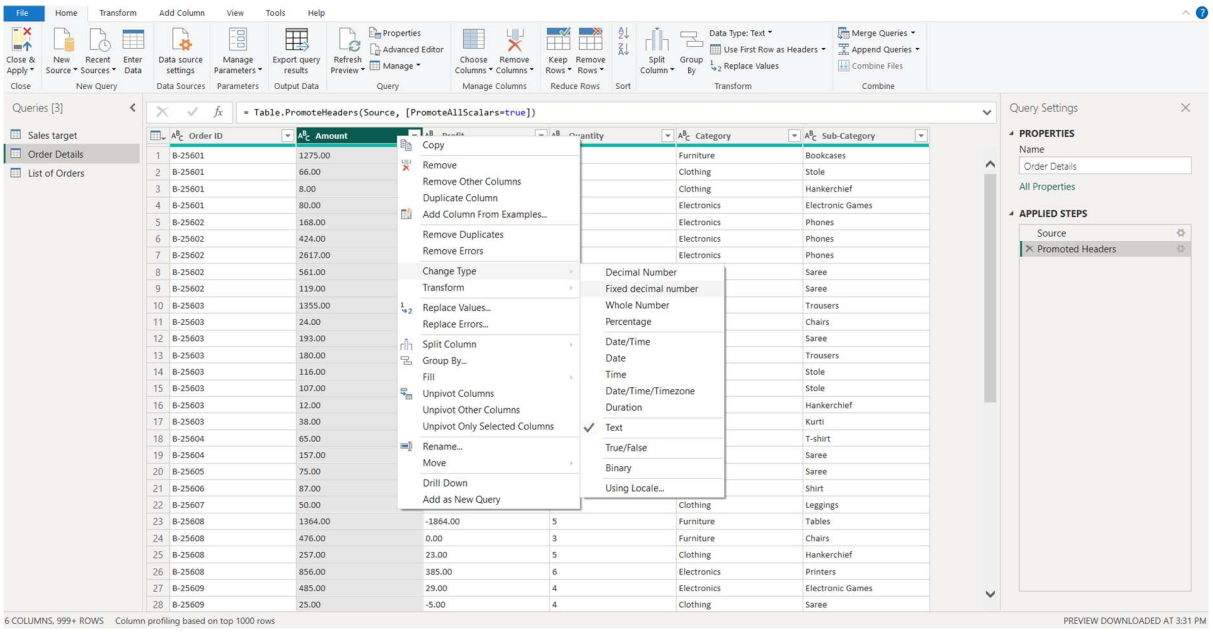


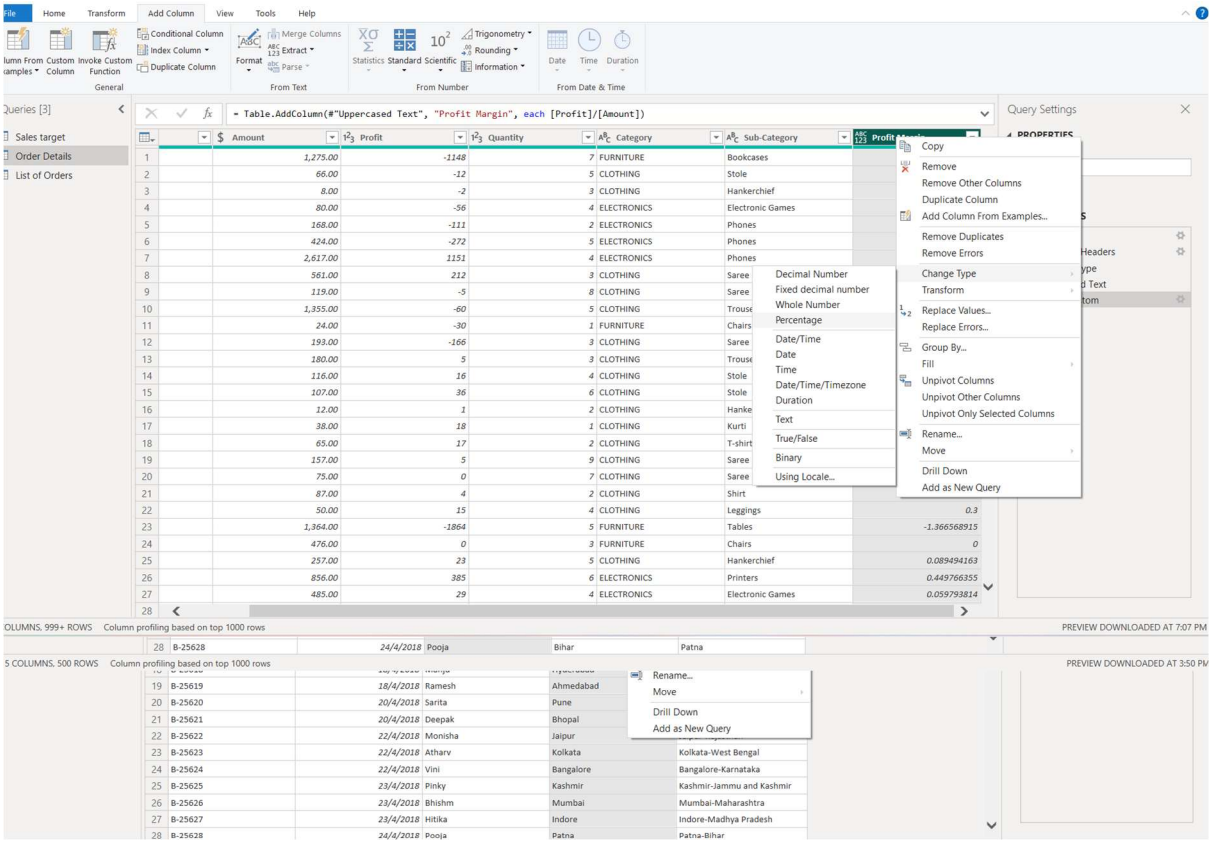
- Set the Order Date column to the Date data type.



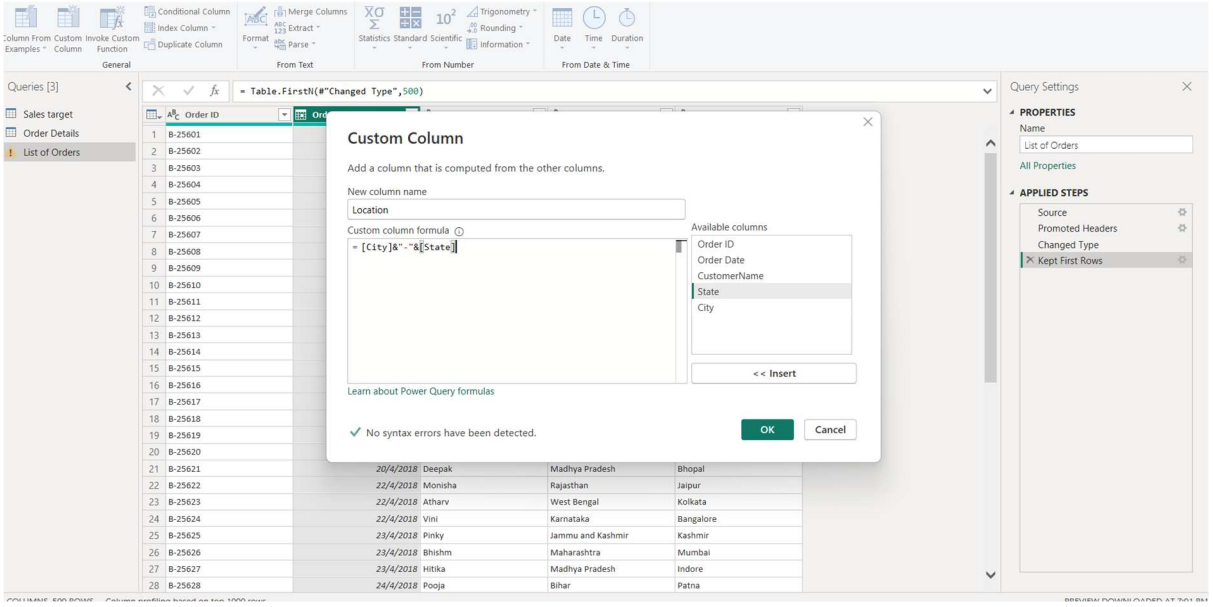
- Change the Amount and Target columns to fixed decimal numbers.



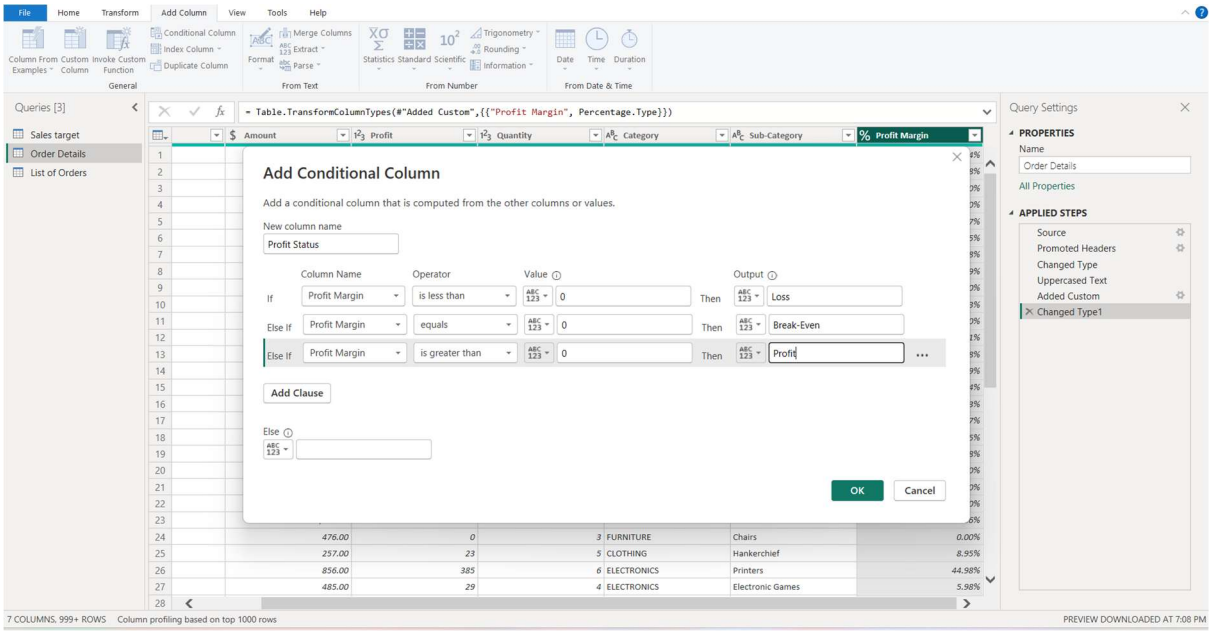




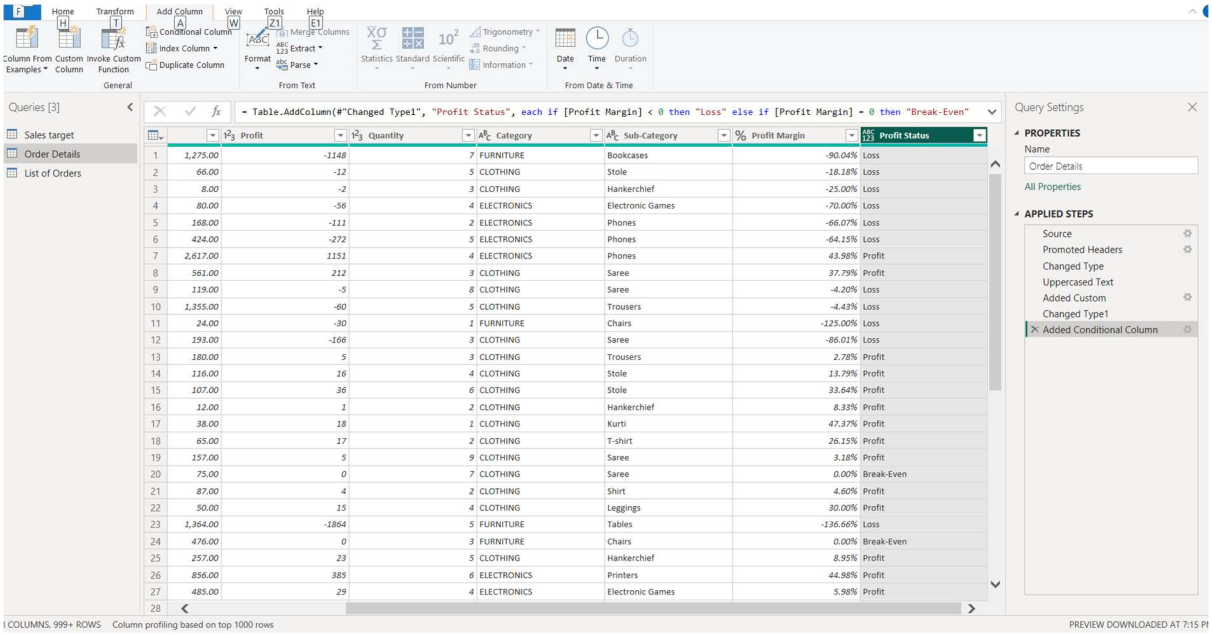
- Merge the City and State columns to create a new column named Location in the format: City, State



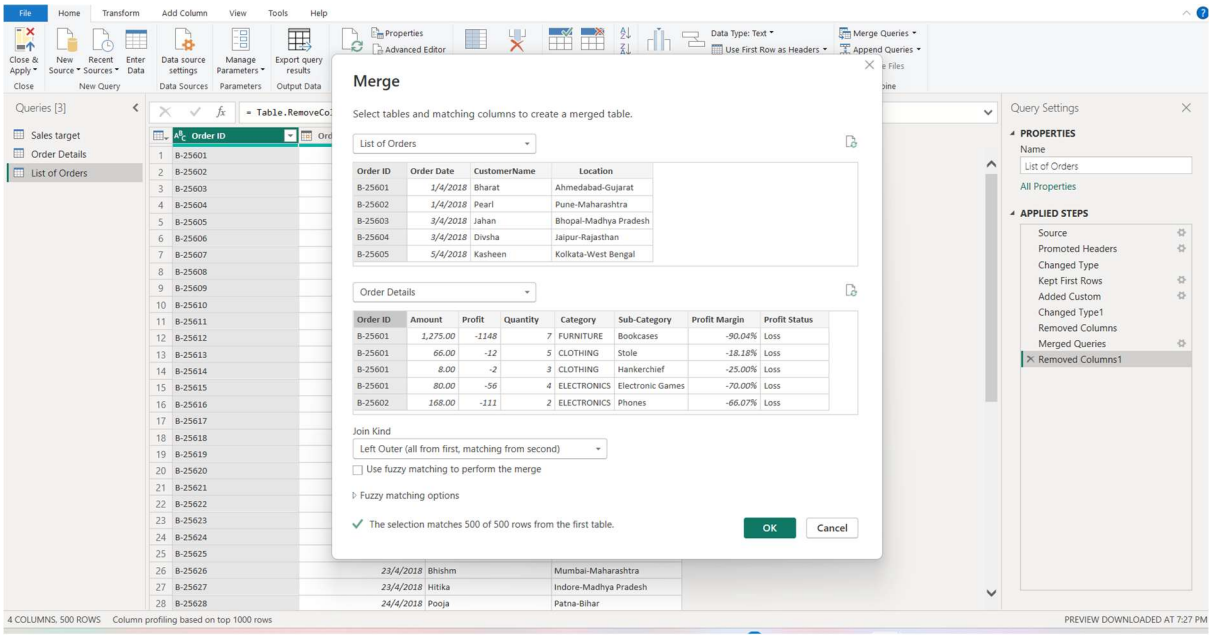
- Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage)

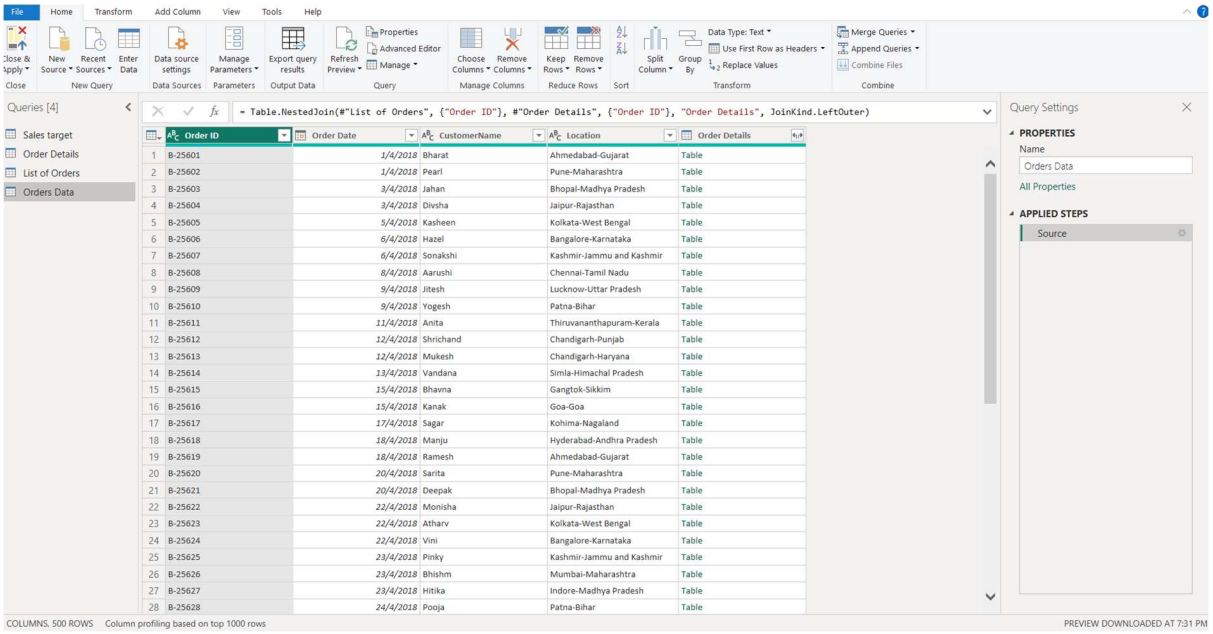


- Add a conditional column named Profit Status based on:

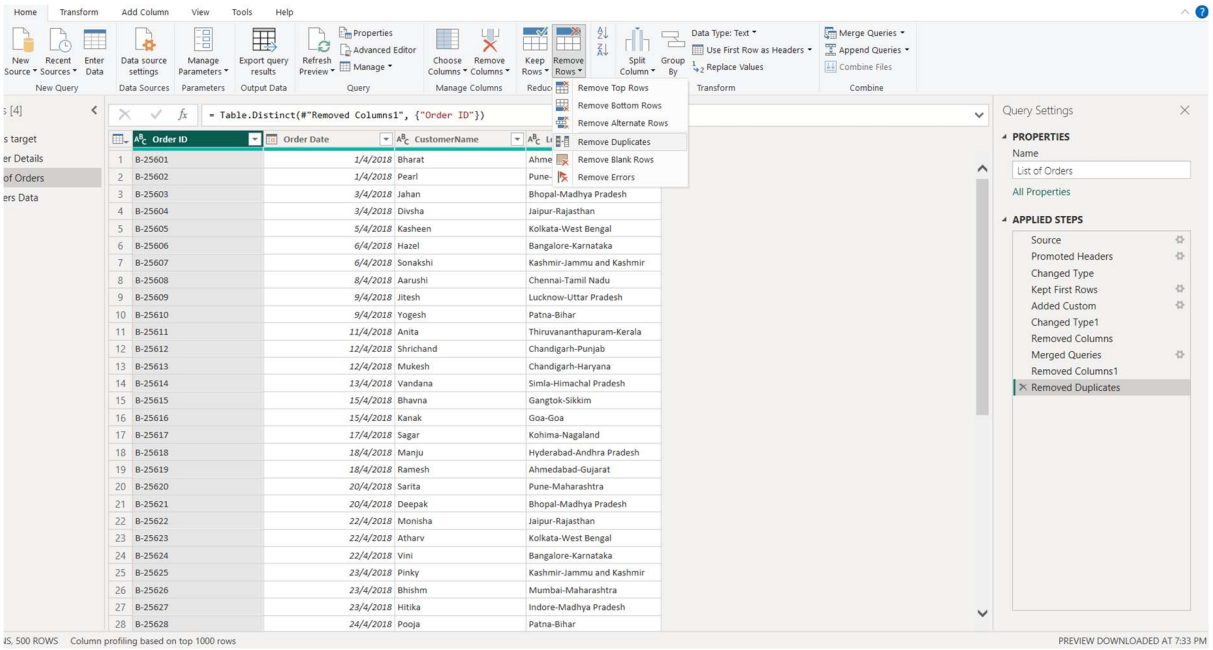


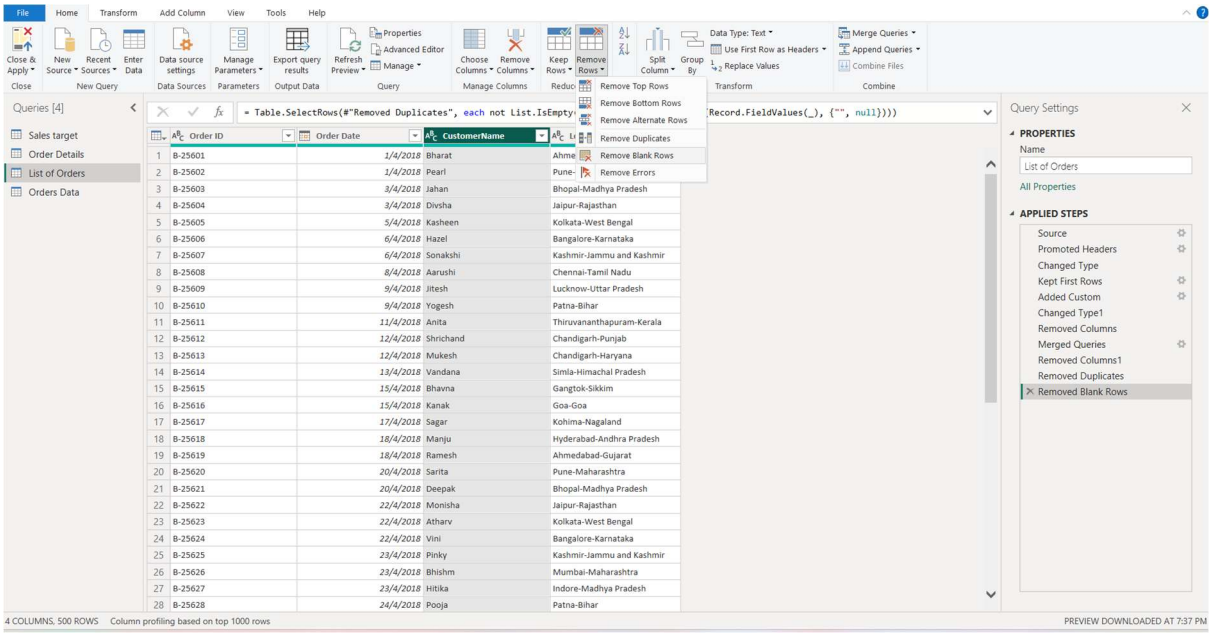
- Merge the List of Orders and Order Details tables using Order ID.
- Name the merged output table as Orders Data.

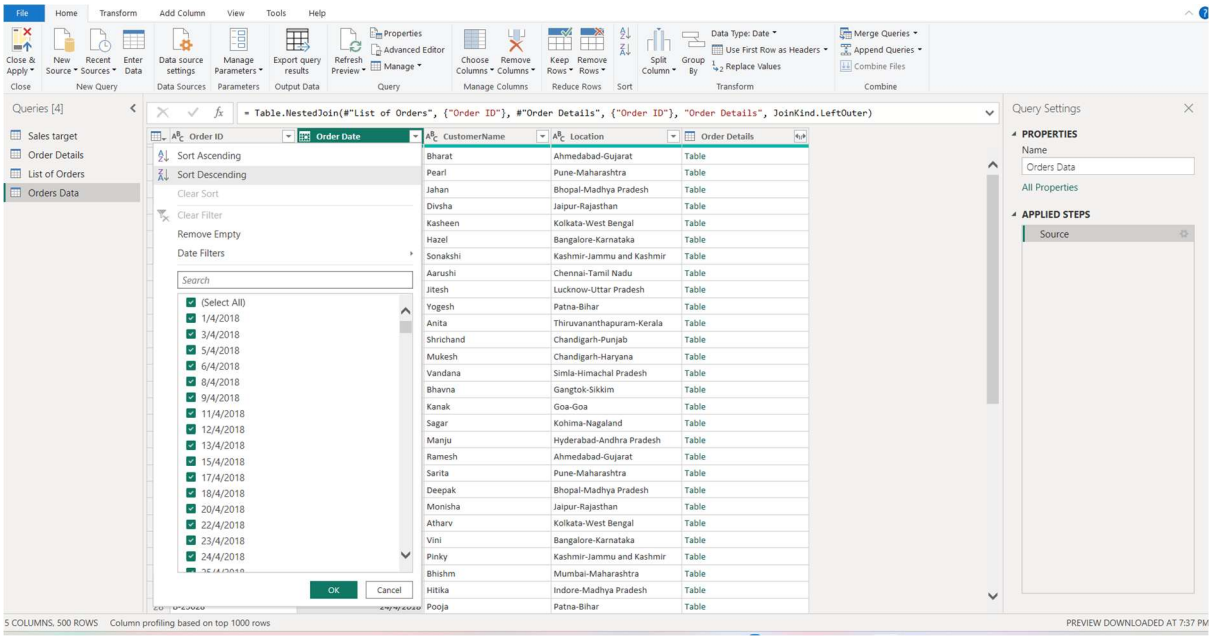




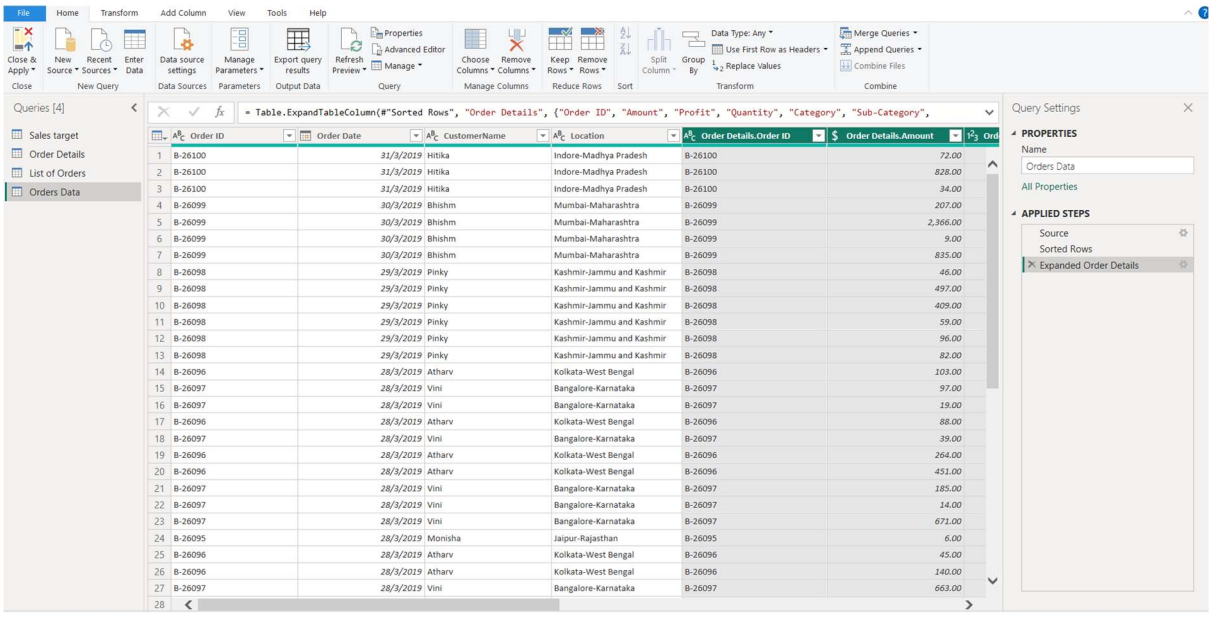
- Replacing missing values
- Removing records



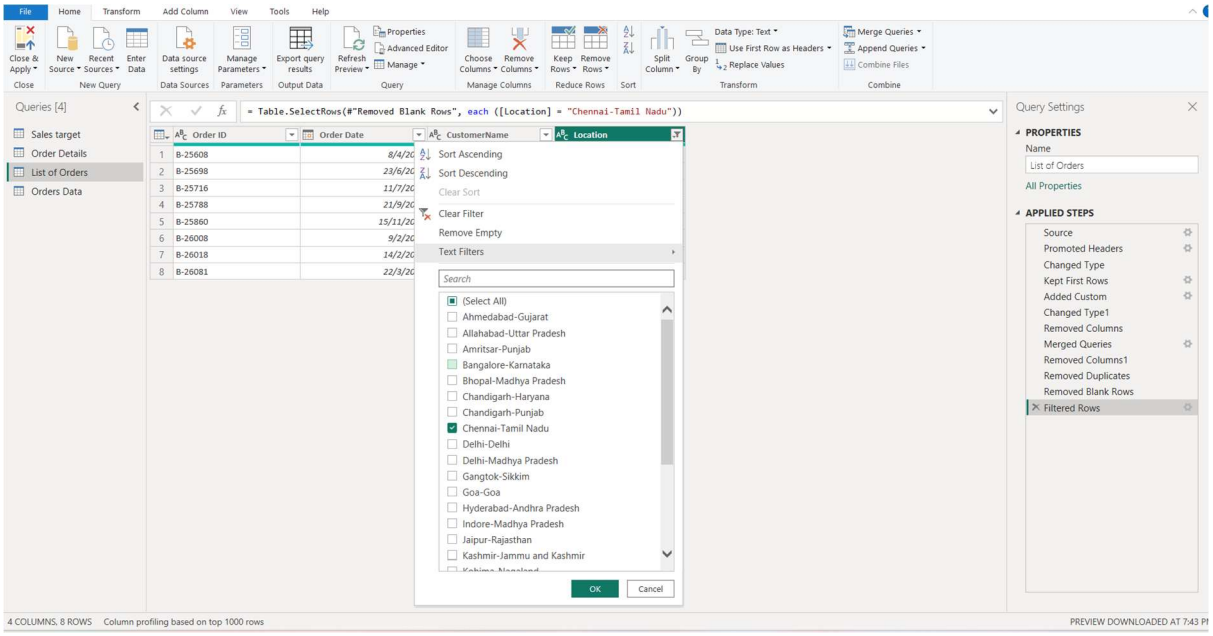


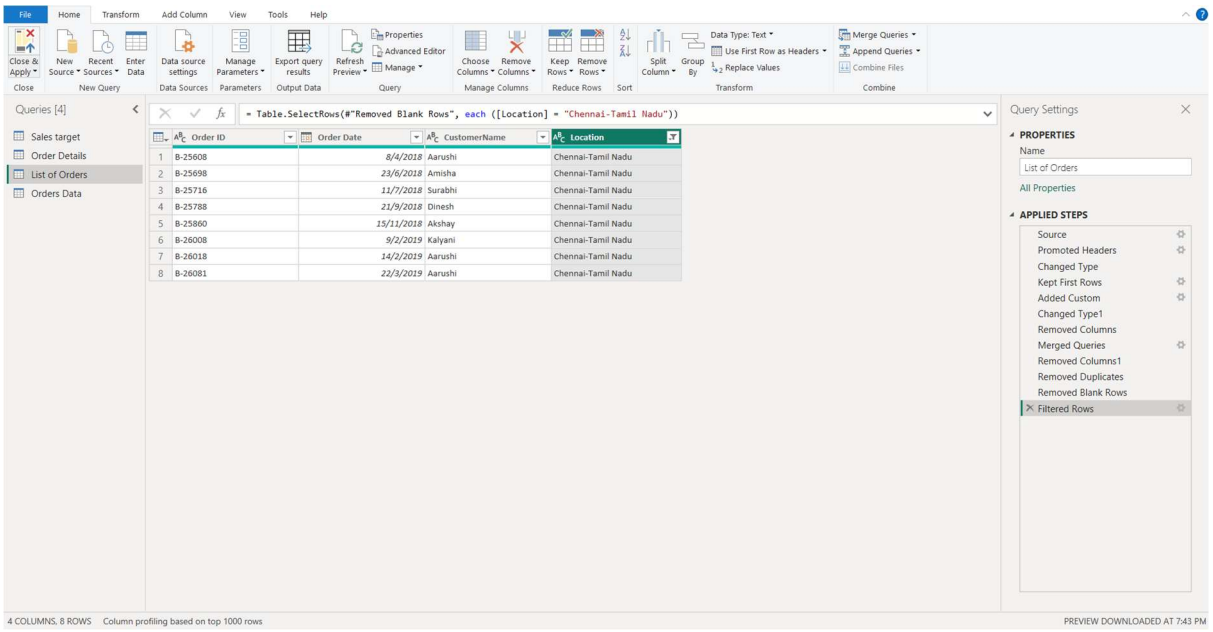


Expanding order Data Table

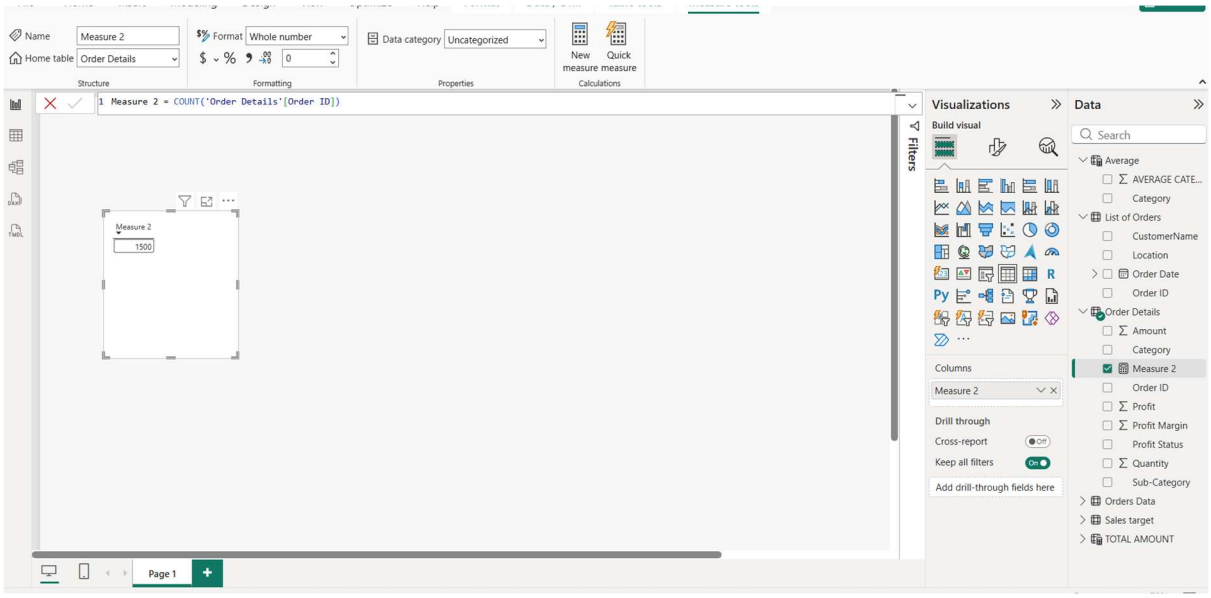


- Apply filters to analyse orders from a specific state (e.g., Tamil Nadu).

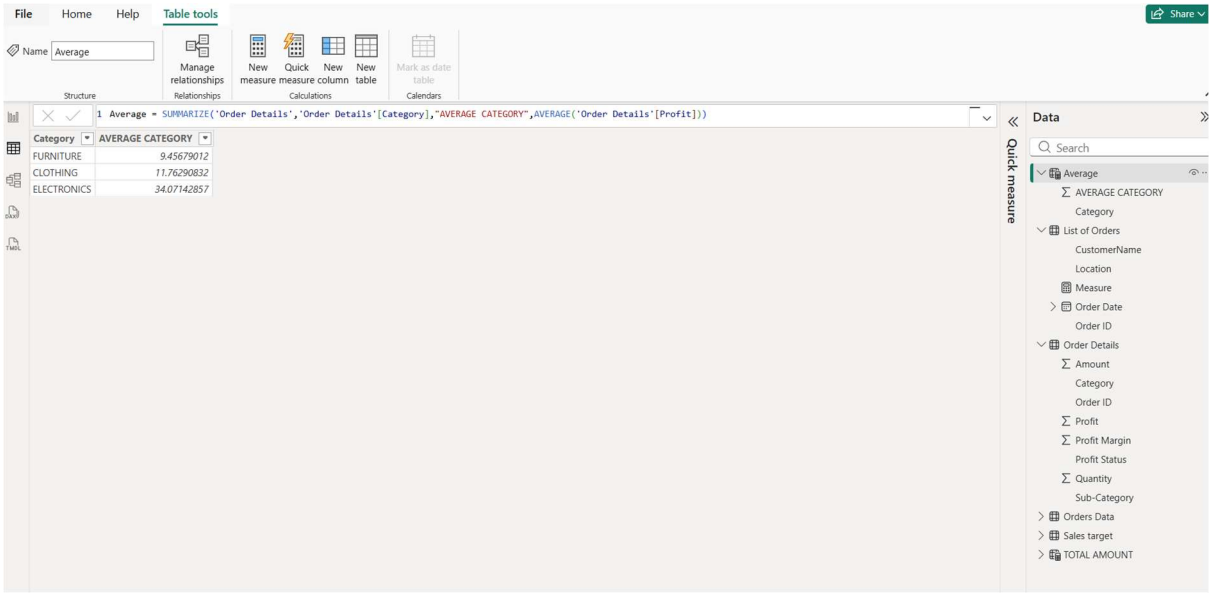




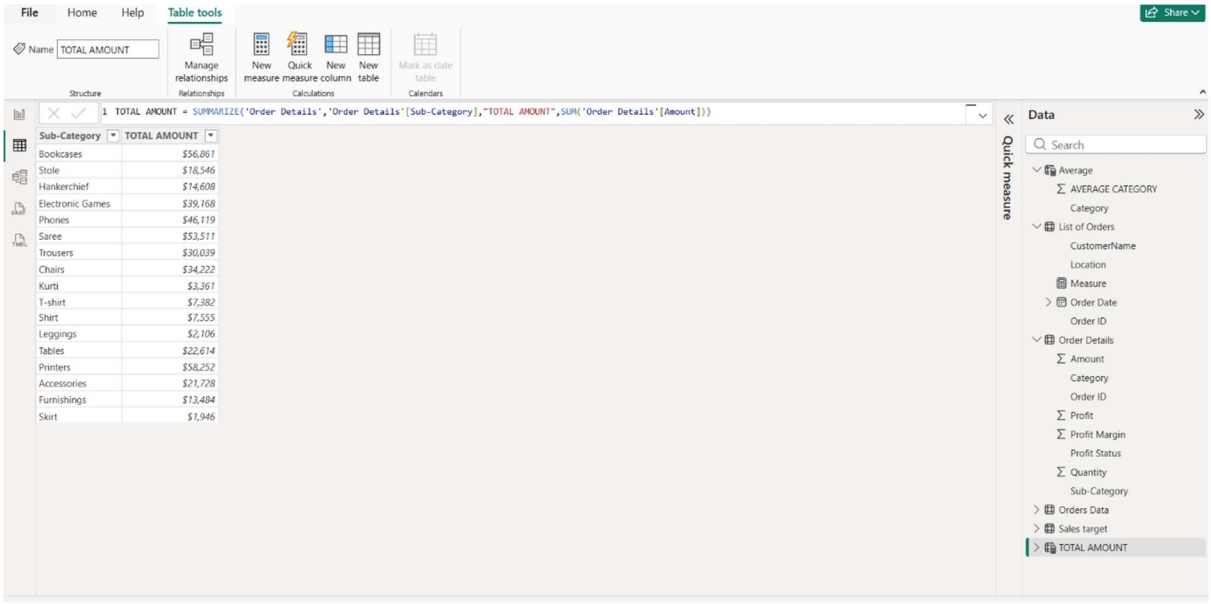
Count of Order ID



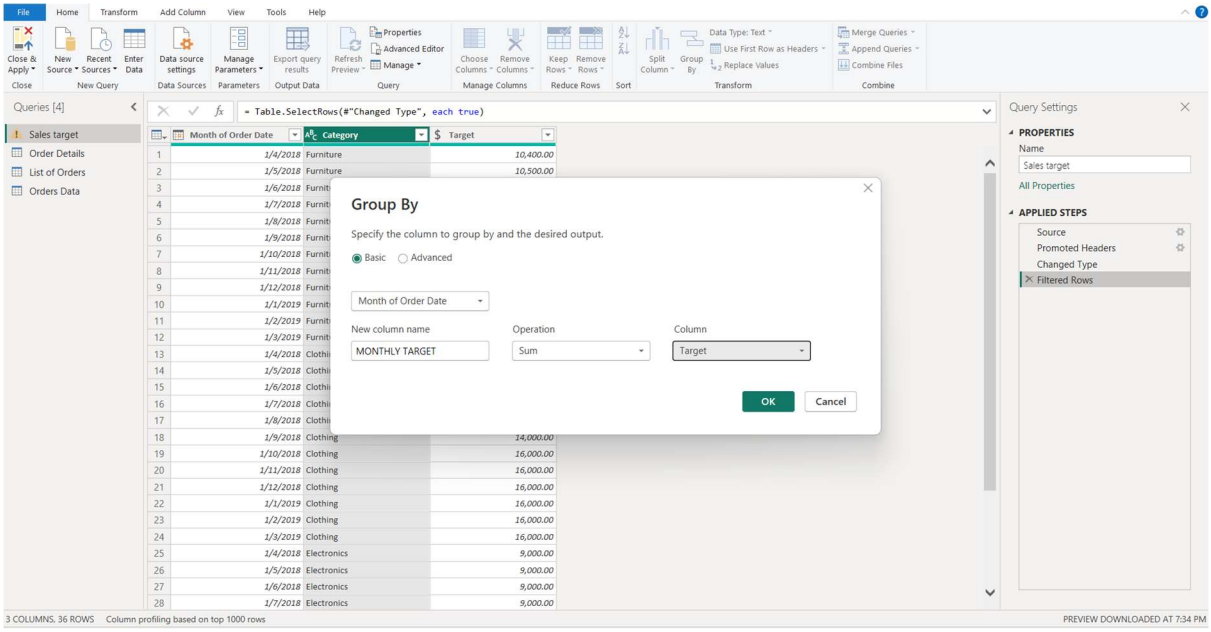
Average Profit by Category



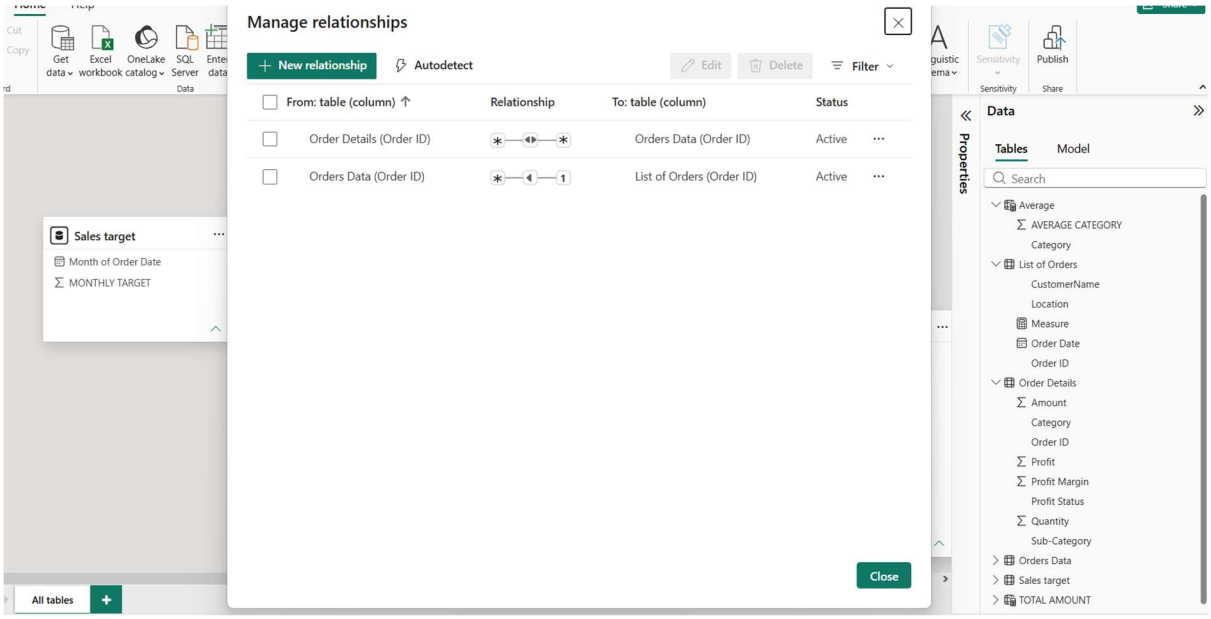
- Total Amount by Sub-Category

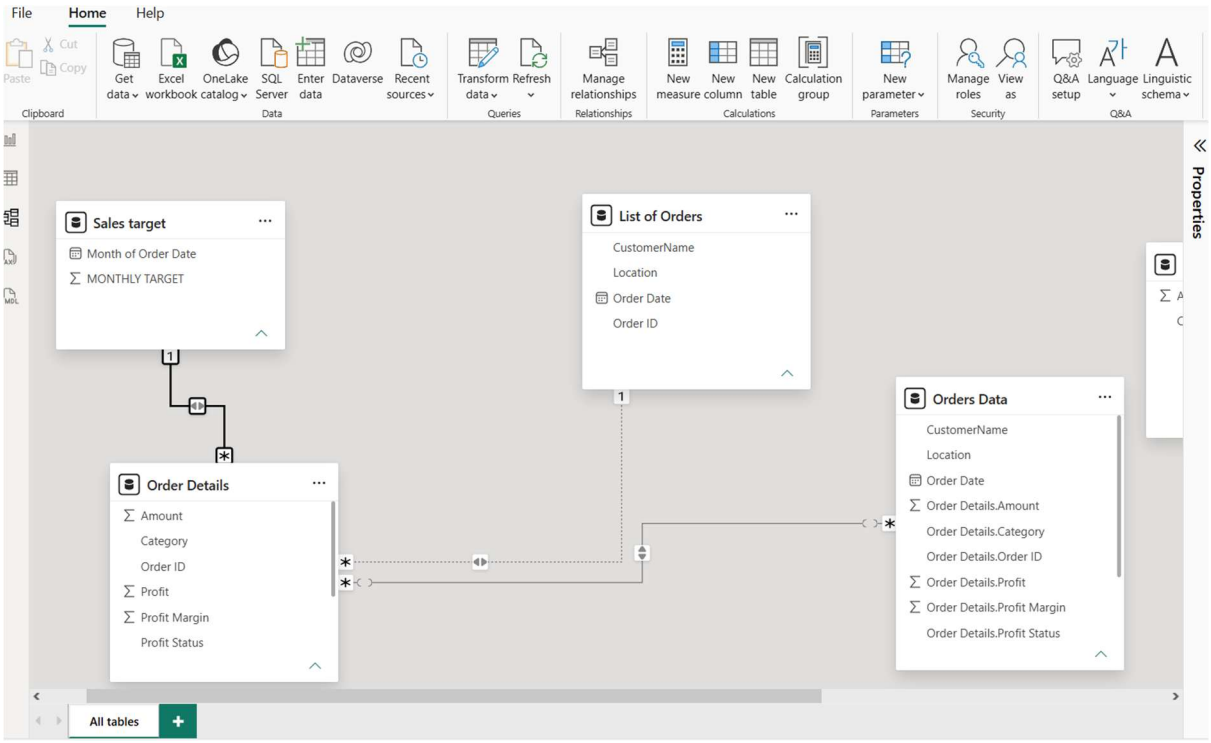


- Aggregate the total Target by Month of order date.



Data Modeling:





CONCLUSION :

This Sales Performance Analysis project uses Excel to analyze order details and monthly sales targets. The analysis helps understand overall sales performance, profitability, product demand, and category-wise performance. Using data cleaning, Excel formulas, and Pivot Tables, the datasets can be transformed into meaningful insights.

The project also helps identify profitable and loss-making categories, compare actual sales with target sales, and recognize areas that require improvement. Based on these insights, businesses can make better decisions, improve sales strategies, increase profitability, and work toward achieving their monthly sales targets.

Overall, this project demonstrates how Microsoft Excel can be used as an effective data analytics tool to convert raw sales data into useful business insights and support data-driven decision-making.

THANKING YOU